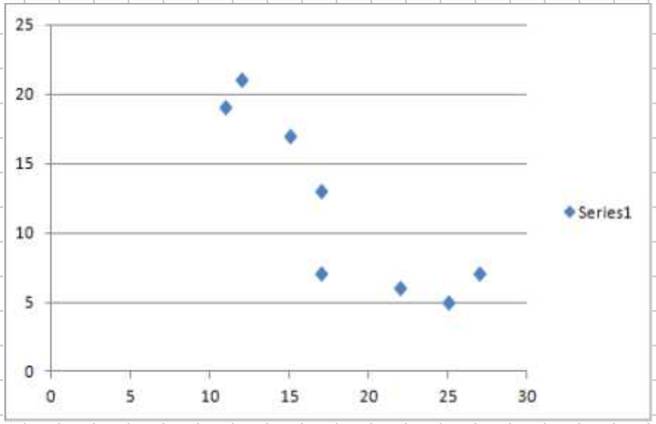
(1) a) Dependent variable : # of crimes (y)
Independent " : # of police (x)

b)

crime



police

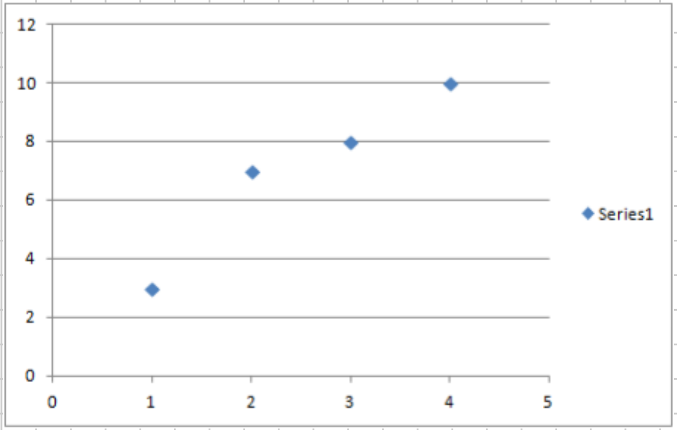
c) $r = -0.8744$

d) There is a strong negative relationship, which shows the importance of increasing the number of police. More police means less crime.

② a) Dependent : sales revenue (y)
Independent : advertising expense (x)

b)

Sales revenue



Advertising expense

c) $r = 0.9648$.

d) $\hat{y} = 1.5 + 2.2x$

e) Intercept : without spending anything on advertising, Haverty's Furniture's sales revenue will be 1.5 mio USD.

slope : when the advertising expense increases by 1 mio USD, the sales revenue increases by 2.2 mio USD.

f) If $x = 3$ mio USD, $\hat{y} = 1.5 + 2.2 \cdot 3 = 8.1$ mio USD.

3 a) $\bar{x} = 33.4$, $\bar{y} = 61.1$

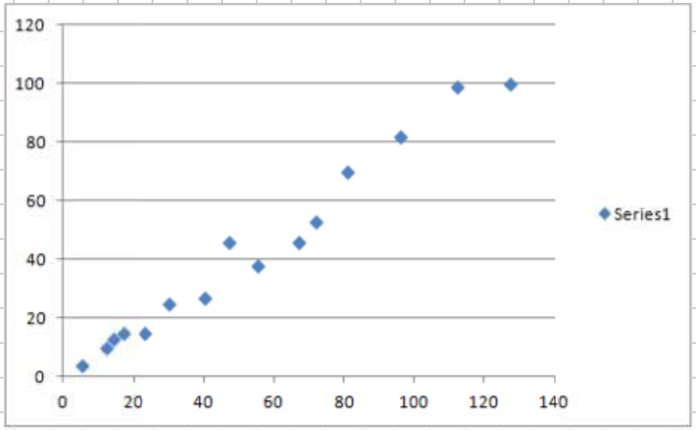
$S_{xx} = 2814.4$, $S_{xy} = 6176.6$, $\hat{\beta} = \frac{6176.6}{2814.4} = 2.195$

$\hat{\alpha} = 61.1 - 2.195 \times 33.4 = -12.201$

$\hat{y} = -12.201 + 2.195x$

b) If $x = 40$, $\hat{y} \approx 75.6 \times 10^3$ USD $= 75{,}600$ USD

4 a)



Yes it does.

b) $\hat{\alpha} \cong -1.1283$, $\hat{\beta} \cong 0.8270$

c) If $x = 50$, $\hat{y} = -1.1283 + 0.8270 \cdot 50 = \underline{\underline{40.2217}}$

d) $\hat{\sigma} = \underline{\underline{5.2405}}$

e) $r^2 = 97.\underline{\underline{53}}\%$

⑤ $\hat{y} = 29.29 - 0.96x$

$n = 8$

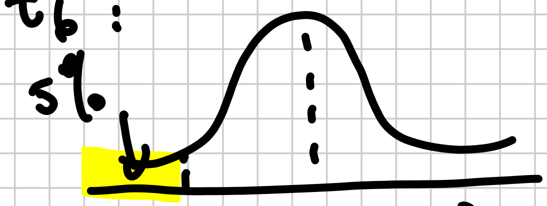
$SE(\hat{\beta}) = 0.22$

1) $H_0: \beta = 0$ vs $H_a: \beta < 0$ $\Rightarrow$ Lower-tailed.

2) $\alpha = 0.05$

3) Test statistic : $T = \dfrac{\hat{\beta}}{SE(\hat{\beta})} \overset{H_0}{\sim} t_6$

4) t_6 :



$t_{6,0.95} = -1.943$

If $t_{obs} < -1.943$, we reject H_0.

5) $t_{obs} = \dfrac{-0.96}{0.22} \simeq -4.36 < -1.943$

$\Rightarrow$ We reject H_0 and conclude that the slope is significantly negative.

(6) $\hat{y} = 1.85 - 0.08x$

$n = 12$

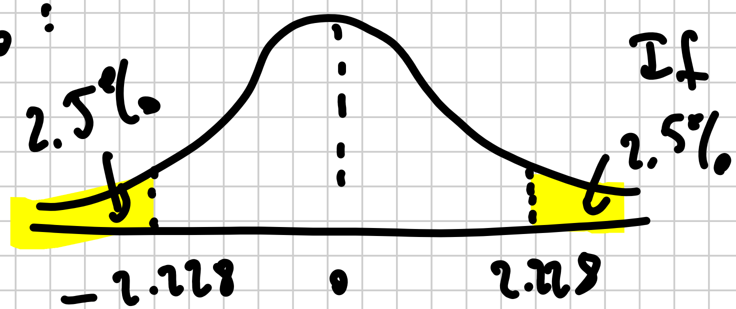
$SE(\hat{\beta}) = 0.03$

1) $H_0: \beta = 0$ vs $H_a: \beta \neq 0$ $\Rightarrow$ Two tailed.

2) $\alpha = 0.05$

3) Test statistic : $T = \dfrac{\hat{\beta}}{SE(\hat{\beta})} \overset{H_0}{\sim} t_{10}$

4) t_{10} :



If $|t_{obs}| > 2.228$, we reject H_0.

5) $t_{obs} = \dfrac{-0.08}{0.03} \simeq -2.67 < -2.228 \Rightarrow$ We reject H_0 and conclude that the slope is significantly different from 0.

7) a) $r = -0.8224$ (using CORREL in Excel)

b) $\hat{\alpha} = 18\,357.74$

$\hat{\beta} = -1533.63$

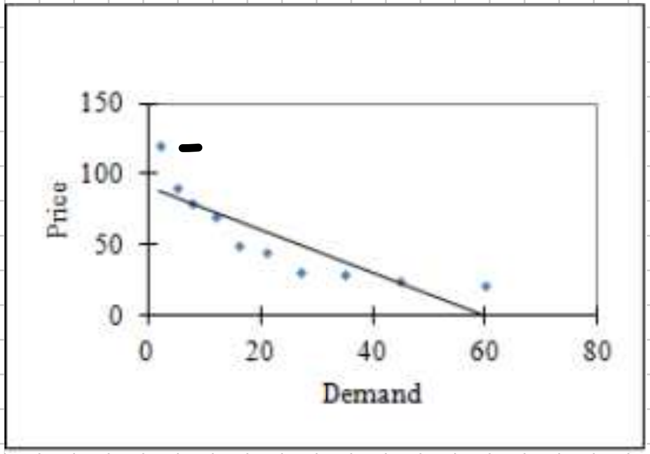
c) $\hat{y} = 18\,357.74 - 1533.63 \cdot 5 = 10\,\underline{\underline{689}}.56$ USD

d) $H_0 : \beta = 0$ vs $H_a : \beta \neq 0$

p-value $= 0.000305 < 0.10$

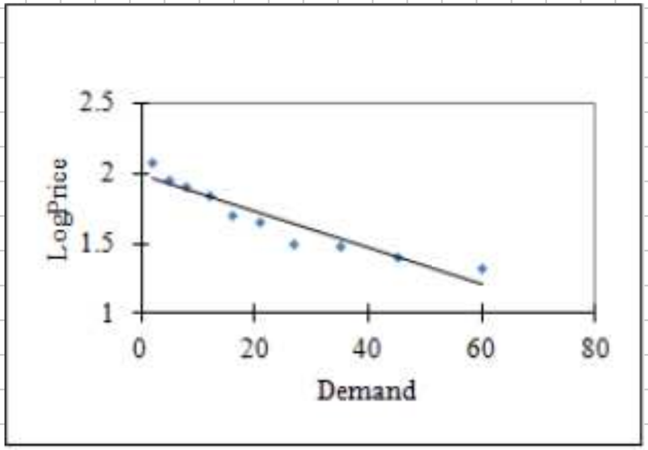
We reject H_0 and conclude that the slope parameter is significant. The age of a fuel-efficient car significantly reduces its price.

8) a) $r = -0.8653$



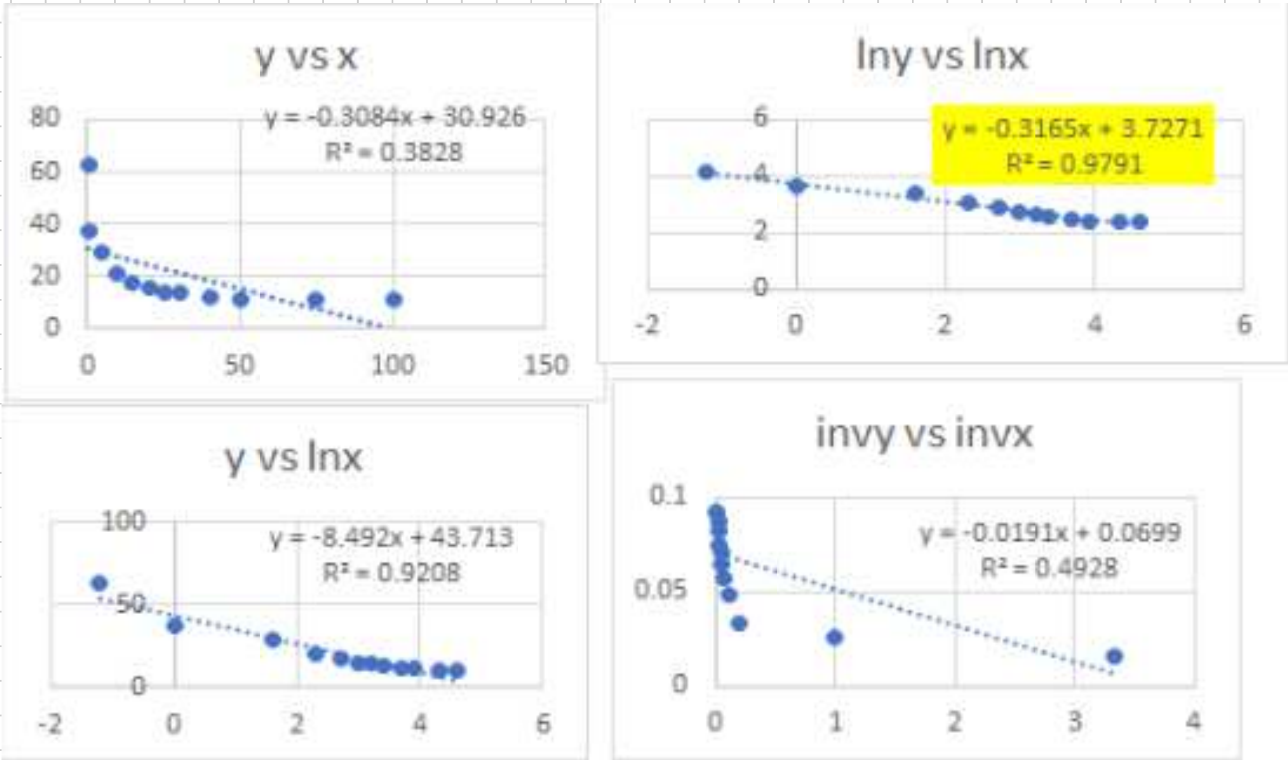
$\Rightarrow$ nonlinear relationship between Price and Demand.

b) $r = -0.946$



$\Rightarrow$ linear relationship between $\lg_{10}$(Price) and Demand is more plausible than between Price and Demand.

9 a)



b) $\ln(y_i) = \alpha + \beta \ln(x_i) + \varepsilon_i$ is the preferred model given the r^2 values in a).

$$\widehat{\ln(y)} = 3.727 - 0.316 \cdot \ln(x)$$

$$x = 45 \rightarrow \widehat{\ln(y)} \simeq 2.524 \rightarrow \hat{y} \simeq \underline{\underline{12.48}}$$

Based on DAD lectures notes and exercices book of V. Czellar, S. Dabo, P. Cogneau, B. Desmarchelier